

technologies designed to strengthen the mission's layered air defence grid. Recently, Israel signed several agreements for the transfer of technology related to defensive systems and offensive weapons.

Defensive technologies include the Arrow system, Rafael's David's Sling, and the Iron Dome. Offensive weapons include Rafael's SPICE-1000 guidance kits, Rampage air-to-ground missiles, Ice Breaker naval cruise missiles, and supersonic Air LORA missiles.

Israel has also expanded collaborative programmes into emerging technologies such as directed-energy systems and laser-based weapons.

Mission Sudarshan Chakra: Key challenges

Implementing Mission Sudarshan Chakra faces several challenges. The biggest challenge lies in integrating multiple systems into a unified national defence grid while operating within constrained timelines, leveraging advanced technologies, coordinating across organisations, and managing complex logistics.

This requires timely completion of activities, availability of critical components, close coordination among the three services and research organisations and industry, sufficient skilled manpower, and sustained investment commitments. Some of the key challenges include:

Systems integration. Diverse radar systems, interceptor platforms, and communication networks are built on different architectures, data formats, latency characteristics, and protocols. Seamless exchange of secured data requires complex middleware, standardised interfaces, and resilient cybersecurity frameworks. The complexity increases further when advanced systems with distinct operational profiles are integrated into the grid. Ensuring the reliability, accuracy, and resilience of integrated systems, even under jamming and electronic warfare conditions, requires a flexible architecture with sufficient redundancy.



Israel's LR SAM/Barak 8 strengthens the layered air defence grid of the mission
(Source: <https://www.facebook.com>)

Flexible design and regular upgrades are crucial for integrating advanced technologies smoothly while maintaining uninterrupted operations.

Human resource. Operating a digitally integrated, multi-layered inter-services network requires highly skilled personnel with expertise in specialised fields such as radar, electronic warfare, cybersecurity, network management, and systems integration. There is a global shortage of such manpower. This shortage affects implementation and can disrupt schedules and operational reliability. Establishing training institutes and developing programmes is time-consuming and requires sustained financial support.

Regular cross-service training programmes, close coordination with research organisations, academia, and industry, along with effective retention of experienced personnel, are necessary to ensure the availability of trained manpower.

Logistics. Nationwide deployment of assets across diverse geographies, ranging from high-altitude radar stations to coastal and island territories, complicates logistics. Each site requires stable power supply, secure communication systems, preventive maintenance, and dependable supply chains to remain operational under adverse conditions. This requires decentralised logistics hubs, regional maintenance facilities, and automated diagnostic systems to sustain uninterrupted operations.

Advanced planning, timely infrastructure availability, and reliable supply chains are essential to support and maintain deployment.

Supply chains. Timely availability of critical components, such as gallium nitride radar modules, high-performance seekers, radiation-hardened processors, and high-density FPGAs, is essential. These components are difficult to procure because of proprietary restrictions